

Multiple Choice (10 marks)

1. The polynomial which results from the expansion of $(x^2+5x+6)^2+(px+q)(x^3+7x^2+3x)$ has degree 2. Find $p+q$.
 - (a) -2
 - (b) -4
 - (c) -1
 - (d) 1
2. How many numbers are in the list $25, 26, 27, \dots, 99, 100$?
 - (a) 75
 - (b) 76
 - (c) 22
 - (d) 23
3. The length of a rectangle is twice its width. Given the length of the diagonal is $5\sqrt{5}$, find the area of the rectangle.
 - (a) 2500
 - (b) 2
 - (c) 50
 - (d) 25
4. Find the remainder when 25^{1059} is divided by 23.
 - (a) 11
 - (b) 13
 - (c) 8
 - (d) 5
5. The sum $1^2 + 2^2 + 3^2 + 4^2 + \dots + n^2 = n(n+1)(2n+1) \div 6$. What is the value of $21^2 + 22^2 + \dots + 40^2$?
 - (a) 41
 - (b) 22140
 - (c) 19270
 - (d) 2870

True / False (10 marks)

Answer True or False

6. True or False: The total number of liters of water in 8 fish tanks, each with 40 liters, is 280 liters.
7. True or False: The solution to the equation $-2k = -34.8$ is $k = -17.4$.
8. True or False: The area of a square with sides equal to the radius of a circle with a circumference of 12 units is 12 square units.
9. True or False: There are 49 students on the dance team who take tap lessons, which is $\frac{4}{9}$ of the total 72 students.
10. True or False: Sapphire can make 25 bouquets if she uses 7 balloons in each bouquet from her 179 balloons.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Marshall has 550 words left to type for his English paper, and he has 25 minutes before class starts. Discuss how you would calculate the minimum typing speed needed to complete his task on time.
12. Discuss the concept of angles in relation to a circle. Specifically, explain how you determine the measure of an angle that represents $\frac{1}{360}$ of a full circle.

End of Paper